i常返 ⟹ i所在互达等价类闭 ⟺ i所在等价类不闭 ⟹ i暂留

i所在互达等价类有限闭 ⟹ i正常返

No.

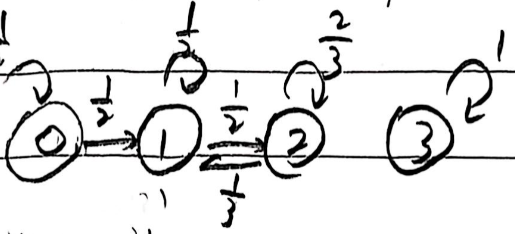
Date.

Homework #3

习题三 10, 16, 17, 20, 21, 23

T10 (1) $P(X_1=1)=P(X_0=0)p_{01}+P(X_0=1)p_{11}=\frac{1}{3}$

$P(X_1=1,X_3=2)=P(X_1=1)p_{12}^{(2)}=\frac{1}{3}\times\frac{7}{12}=\frac{7}{36}$



$P(X_2=1)=P(X_0=0)p_{01}^{(2)}+P(X_0=1)p_{11}^{(2)}$

$=\frac{1}{3}\times\frac{1}{2}+\frac{1}{3}\times\frac{5}{12}=\frac{11}{36}$

$P(X_{10}=0)=P(X_0=0)p_{00}^{(10)}=\frac{1}{3}\times(\frac{1}{2})^{10}=\frac{1}{3072}$

(2) $f_{00}^{(1)}=\frac{1}{2}$, $n\geq 2$, $f_{00}^{(n)}=0$, $f_{00}=\frac{1}{2}$, 0是暂留态,

$1\leftrightarrow 2$, 且$\{1,2\}$为有限闭集, 故1,2为正常返态

对1,2, $P=\begin{bmatrix}\frac{1}{2} & \frac{1}{2}\\ \frac{1}{3} & \frac{2}{3}\end{bmatrix}$

则 $\begin{cases}\pi_1=\frac{1}{2}\pi_1+\frac{1}{3}\pi_2\\ \pi_2=\frac{1}{2}\pi_1+\frac{2}{3}\pi_2\\ \pi_1+\pi_2=1\end{cases}\Rightarrow\begin{cases}\pi_1=\frac{2}{5}\\ \pi_2=\frac{3}{5}\end{cases}\Rightarrow\begin{cases}\mu_1=\frac{5}{2}\\ \mu_2=\frac{5}{3}\end{cases}$

$f_{33}^{(1)}=1$, $f_{33}=1$, $\mu_3=1$, 3是正常返态,

{Xn}不可约 ① {Xn}存在平稳分布 ⟺ {Xn}正常返

② {Xn}正常返，$\pi_j=\frac{1}{\mu_j}$

③ {Xn}遍历，$\lim p_{ij}^{(n)}=\pi_j$（周期不讨论）

T16 0是暂留态，$\lim\limits_{n\to\infty}P(X_n=0)=0$

$\lim\limits_{n\to\infty}P(X_n=3)=P(X_0=3)=\frac{1}{3}$

$\lim\limits_{n\to\infty}P(X_n=1)=\lim\limits_{n\to\infty}P(X_0=0)P(X_n=1|X_0=0)+\lim\limits_{n\to\infty}P(X_0=1)P(X_n=1|X_0=1)$

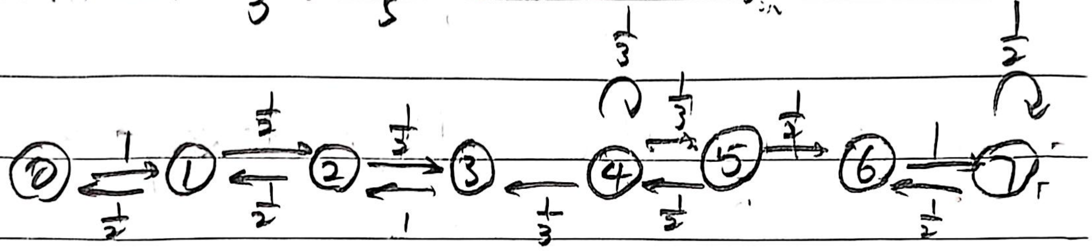
$\lim\limits_{n\to\infty}P(X_n=1|X_0=0)=\frac{1}{2}\lim\limits_{n\to\infty}P(X_n=1|X_1=0)+\frac{1}{2}\lim\limits_{n\to\infty}P(X_n=1|X_1=1)$

故 $\lim\limits_{n\to\infty}P(X_n=1|X_0=0)=\lim\limits_{n\to\infty}P(X_n=1|X_1=1)=\pi_1$

∴ $\lim\limits_{n\to\infty}P(X_n=1)=\frac{2}{3}\pi_1=\frac{4}{15}$

$\lim\limits_{n\to\infty}P(X_n=2)=\frac{2}{3}\pi_2=\frac{2}{5}$ $\lim\limits_{n\to\infty}P(X_n=3)=\frac{1}{3}$

T17



(1) 互达等价类{0,1,2,3}，{4,5}，{6,7}

闭的互达等价类{0,1,2,3}，{6,7}

(2) $d(0)=d(1)=d(2)=d(3)=2$，0,1,2,3为正常返态，

$d(4)=d(5)=1$，4,5为暂留态

$d(6)=d(7)=1$，6,7为正常返态

对{0,1,2,3}，

$$P=\begin{pmatrix}0&1&0&0\\ \frac{1}{2}&0&\frac{1}{2}&0\\ 0&\frac{1}{2}&0&\frac{1}{2}\\ 0&0&1&0\end{pmatrix} \Rightarrow \begin{cases}\pi_0=\frac{1}{2}\pi_1\\ \pi_1=\pi_0+\frac{1}{2}\pi_2 \Rightarrow\\ \pi_2=\frac{1}{2}\pi_1+\pi_3\\ \pi_3=\frac{1}{2}\pi_2\\ \pi_1+\pi_2+\pi_3+\pi_4=1\end{cases} \begin{cases}\pi_0=\frac{1}{6}\\ \pi_1=\frac{1}{3}\\ \pi_2=\frac{1}{3}\\ \pi_3=\frac{1}{6}\end{cases} \Rightarrow \begin{cases}\mu_0=6\\ \mu_1=3\\ \mu_2=3\\ \mu_3=6\end{cases}$$

④ 暂留/零常返 $\forall i, \lim\limits_{n\to\infty} p_{ij}^{(n)} = 0$

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对$\{6,7\}$ $P = \begin{array}{c|cc} & 6 & 7 \\ \hline 6 & 0 & 1 \\ 7 & \frac{1}{2} & \frac{1}{2} \end{array} \Rightarrow \begin{cases} \pi_6 = \frac{1}{2}\pi_7 \\ \pi_6 + \pi_7 = 1 \end{cases} \Rightarrow \begin{cases} \pi_6 = \frac{1}{3} \\ \pi_7 = \frac{2}{3} \end{cases} \Rightarrow \begin{cases} \mu_6 = 3 \\ \mu_7 = \frac{3}{2} \end{cases}$

(3) $\lim\limits_{n\to\infty} p_{45}^{(n)} = 0 \quad \lim\limits_{n\to\infty} p_{67}^{(n)} = \pi_7 = \frac{2}{3}$

(4) $\lim\limits_{n\to\infty} p(X_n=4) = \lim\limits_{n\to\infty} p(X_n=5) = 0$

$\lim\limits_{n\to\infty} p(X_n=6 \mid X_0=4) = \frac{1}{3}\lim\limits_{n\to\infty} p(X_n=6 \mid X_1=4) + \frac{1}{3}\lim\limits_{n\to\infty} p(X_n=6 \mid X_1=5)$

$= \frac{1}{3}\lim\limits_{n\to\infty} p(X_n=6 \mid X_1=4) + \frac{1}{6}\lim\limits_{n\to\infty} p(X_n=6 \mid X_2=4) + \frac{1}{6}\pi_6$

故 $\lim\limits_{n\to\infty} p(X_n=6 \mid X_0=4) = \frac{1}{3}\pi_6 = \frac{1}{9}$

$\lim\limits_{n\to\infty} p(X_n=6) = \frac{1}{2}\lim\limits_{n\to\infty} p(X_n=6 \mid X_0=4) = \frac{1}{18}$

同理 $\lim\limits_{n\to\infty} p(X_n=7 \mid X_0=4) = \frac{1}{3}\pi_7 = \frac{2}{9}$

$\lim\limits_{n\to\infty} p(X_n=7) = \frac{1}{2}\lim\limits_{n\to\infty} p(X_n=7 \mid X_0=4) = \frac{1}{9}$

T20 $I = \{1,2,3,4,5,6,7,8,9\}$

$$P = \begin{array}{c|ccccccccc} & 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \\ \hline 1 & 0 & \frac{1}{2} & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 \\ 2 & \frac{1}{3} & 0 & \frac{1}{3} & 0 & \frac{1}{3} & 0 & 0 & 0 & 0 \\ 3 & 0 & \frac{1}{2} & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 \\ 4 & \frac{1}{3} & 0 & 0 & 0 & \frac{1}{3} & 0 & \frac{1}{3} & 0 & 0 \\ 5 & 0 & \frac{1}{4} & 0 & \frac{1}{4} & 0 & \frac{1}{4} & 0 & \frac{1}{4} & 0 \\ 6 & 0 & 0 & \frac{1}{3} & 0 & \frac{1}{3} & 0 & 0 & 0 & \frac{1}{3} \\ 7 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 8 & 0 & 0 & 0 & 0 & \frac{1}{3} & 0 & \frac{1}{3} & 0 & \frac{1}{3} \\ 9 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{array}$$

$h_i = P(T_7 < \infty \mid X_0 = i)$

则 $h_2 = h_5 = h_8 = \frac{1}{2}$，$h_7 = 1$，$h_9 = 0$

$$\begin{cases} h_1 = \frac{1}{2}h_2 + \frac{1}{2}h_4 \\ h_4 = \frac{1}{3}h_1 + \frac{1}{3}h_5 + \frac{1}{3}h_7 \end{cases} \Rightarrow h_1 = \frac{3}{5}$$

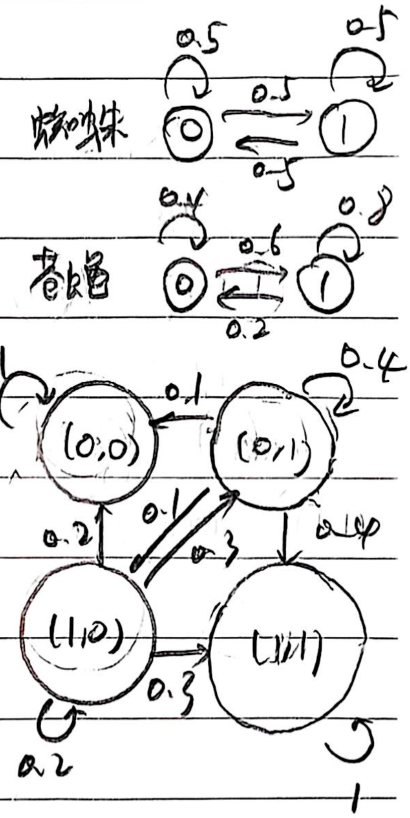
T2 (1) $P\{Z_{n+1} = (X_{n+1}, Y_{n+1}) \mid Z_n = (X_n, Y_n), Z_{n-1} = (X_{n-1}, Y_{n-1}), \dots, Z_0 = (X_0, Y_0)\}$

$= P\{Z_{n+1} = (X_{n+1}, Y_{n+1}) \mid Z_n = (X_n, Y_n)\}$ 且与 n 无关

故 $\{Z_n\}$ 为时齐马氏链

$I = \{(0,0), (0,1), (1,0), (1,1)\}$

$$P = \begin{array}{c} \\ (0,0) \\ (0,1) \\ (1,0) \\ (1,1) \end{array} \begin{array}{c} \begin{array}{cccc} (0,0) & (0,1) & (1,0) & (1,1) \end{array} \\ \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0.1 & 0.4 & 0.1 & 0.4 \\ 0.2 & 0.3 & 0.2 & 0.3 \\ 0 & 0 & 0 & 1 \end{pmatrix} \end{array}$$



(2) $h_{(i,j)} = P(T_{(0,0)} < \infty \mid Z_0 = (i,j))$

$h_{(0,0)} = 1$，$h_{(1,1)} = 0$

$$\begin{cases} h_{(0,1)} = 0.1h_{(0,0)} + 0.4h_{(0,1)} + 0.1h_{(1,0)} + 0.4h_{(1,1)} \\ h_{(1,0)} = 0.2h_{(0,0)} + 0.3h_{(0,1)} + 0.2h_{(1,0)} + 0.3h_{(1,1)} \end{cases}$$

$\Rightarrow h_{(0,1)} = \frac{2}{9}$

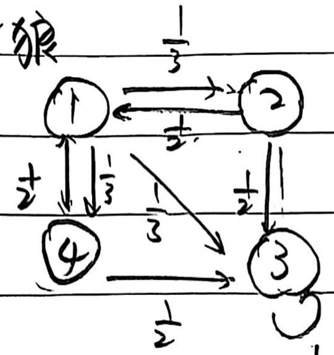
(3) $a(i,j) = E(T_C \mid \xi_0 = (i,j))$ $C = \{(0,0), (1,1)\}$

$a(0,0) = 0$, $a(1,1) = 0$

$$\begin{cases} a(0,1) = 1 + 0.1a(1,0) + 0.4a(0,1) \\ a(1,0) = 1 + 0.2a(1,0) + 0.3a(0,1) \end{cases} \Rightarrow a(0,1) = 2$$

T23 (1) $P(X_1=4, X_3=2)$

$= P_{14}P_{42}^{(2)} = \frac{1}{3} \times \frac{1}{2} \times \frac{1}{3} = \frac{1}{18}$

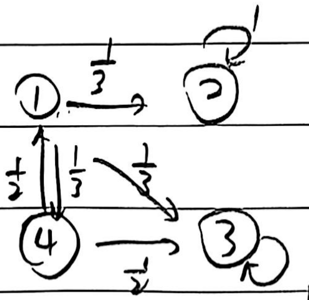


(2) $P(X_2=1 \mid X_3=3)$

$$= \frac{P(X_2=1, X_3=3)}{P(X_3=3)} = \frac{(\frac{1}{3}\times\frac{1}{2}+\frac{1}{2}\times\frac{1}{3})\times\frac{1}{3}}{(\frac{1}{3}\times\frac{1}{2}+\frac{1}{2}\times\frac{1}{3})\times\frac{1}{3}+\frac{1}{3}\times\frac{1}{2}+\frac{1}{3}\times\frac{1}{2}+\frac{1}{3}} = \frac{1}{7}$$

(3) $$f_{12}^{(n)} = \begin{cases} 0 & n\text{为偶数} \\ \frac{1}{3}\times(\frac{1}{6})^{\frac{n-1}{2}} & n\text{为奇数} \end{cases}$$

(4) $h_i = P(T_2 < \infty \mid X_0 = i)$



$h_2 = 1$, $h_3 = 0$

$$\begin{cases} h_1 = \frac{1}{3} + \frac{1}{3}h_4 \\ h_4 = \frac{1}{2}h_1 \end{cases} \Rightarrow h_1 = \frac{2}{5}$$

(5) $a_i = E(T_3 \mid X_0 = i) \Rightarrow \begin{cases} a_1 = 1 + \frac{1}{3}a_2 + \frac{1}{3}a_4 \\ a_2 = 1 + \frac{1}{2}a_1 \\ a_4 = 1 + \frac{1}{2}a_1 \end{cases} \Rightarrow a_1 = \frac{5}{2}$

$a_3 = 0$